

Experiment 1: Caesar Cipher

Program:

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
#include <string.h>
```

```
void encrypt(char* message, int key) {  
    for (int i = 0; message[i] != '\0'; ++i) {  
        char ch = message[i];  
  
        if (isalpha(ch)) {  
            char base = islower(ch) ? 'a' : 'A';  
            ch = (ch - base + key) % 26 + base;  
        }  
  
        message[i] = ch;  
    }  
}
```

```
int main() {  
    char message[1000];  
    int key;  
  
    printf("Enter a message: ");  
    fgets(message, sizeof(message), stdin);
```

```
printf("Enter key (1-25): ");  
scanf("%d", &key);  
  
if (key < 1 || key > 25) {  
    printf("Invalid key! Must be between 1 and 25.\n");  
    return 1;  
}  
  
encrypt(message, key);  
  
printf("Encrypted message: %s\n", message);  
  
return 0;  
}
```

Input and Output :

```
Enter a message: HELLO  
Enter key (1-25): 3  
Encrypted message: KHOOR
```